COURSE TITLE	**VLSI DESIGN**
COURSE CODE	**01CT1514**
COURSE CREDITS	**4**

Objective:

1 The objective of this course is to introduce students with various concepts and methods of digital system design techniques. To acquire knowledge of MOS Transistor, understand CMOS Fabrication process and learn the design of CMOS Logic circuits and subsystems. Learning this course would improve the employment potential of students in VLSI and Semiconductor industry.

Course Outcomes: After completion of this course, student will be able to:

1 Acquire basics of VLSI design flow, design methodologies, design quality, design for manufacturability and testability (Apply)

2 Develop fundamentals in fabrication of MOSFETs and MOS transistor (Understand)

3 Analyze MOS inverters with its static and switching characteristics (Analyze)

4 Analyze combinational and sequential MOS logic circuit (Analyze)

5 Acquire basic concepts of low power CMOS logic circuits (Apply)

Pre-requisite of course: Elementary knowledge about Electronics including some experience of circuit designing and logic development. Fundamentals of Computer Programming & Utilization. Basic knowledge of digital electronics and real-life digital systems.

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
3	0	2	50	30	20	25	25

Contents : Unit	Topics	Contact Hours
1	**Introduction** Overview of VLSI design methodology, VLSI design flow, Design hierarchy,, Concept of regularity, Modularity, and Locality, VLSI design style, Design quality, package technology, introduction to FPGA and CPLD, computer aided design technology.	3
2	**Fabrication of MOSFET** Introduction, Fabrication Process flow: Basic steps, Wafer formation, , Photolithography, Gate oxide, gate and drain formation and Contacts and Metallization, nMOS Fabrication. C-MOS n-Well Process, Layout Design rules, full custom mask layout design.	5

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Contents : Unit	Topics	Contact Hours
3	**MOS Transistor Theory** Introduction and I-V Characteristics, Non ideal I-V effects: Velocity saturation, mobility degradation and Channel length modulation, Non ideal I-V effects: Body Effect, Sub threshold conduction, Junction leakage, MOSFET scaling and small geometry effects.	8
4	**MOS Inverters** Introduction, Resistive load inverter, Inverters with n-type MOSFET load, CMOS Inverter, Propagation delay, Interconnect delay, switching power dissipation of inverter	7
5	**Combinational and Sequential CMOS Logic Circuits** MOS logic circuit with depletion load, CMOS logic circuits, Complex logic circuit, pass transistor circuits, CMOS with transmission gate, Multiplexer and XOR gate using CMOS transmission gate, Level sensitive latches and edge triggered flip flops	8
6	**Chip I/P and O/P Circuits** ESD protection, Input circuits, Output circuits, On chip Clock Generation and Distribution, Latch –Up and its Prevention	2
7	**Low-power CMOS logic circuits** Introduction, Overview of Power Consumption, Low power design, Leakage power dissipation, Reduction in switching activity, Gated clock signals	4
8	**Design for manufacturability** Introduction, Process variations, Basic concepts and definitions	2
9	**Design for Testability** Introduction, Fault types and models, Controllability and observability,, Various techniques to identify the faults.	3
Total Hours		**42**

Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
1	**Experiments 1** Introduction to SPICE tool.	2
2	**Experiments 2** Develop the current-voltage characteristics of NMOS transistor	2
3	**Experiments 3** Develop the current-voltage characteristics of PMOS transistor	2
4	**Experiments 4** Implement NAND gate using CMOS transistors and simulate the design.	2
5	**Experiments 5** Implement NOR gate using CMOS transistors and simulate the design.	2

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Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
6	**Experiments 6** CMOS implementation of XOR and XNOR gate. Simulate the design.	2
7	**Experiments 7** Draw CMOS inverter and simulate the design with input and output waveforms	2
8	**Experiments 8** Draw the layout of NAND and NOR gate and simulate the design	2
9	**Experiments 9** Draw the layout of AND and OR gate and simulate the design.	2
10	**Experiments 10** Implement the 2x1 MUX using Transmission gate and simulate the design.	2
11	**Experiments 11** Develop layout of CMOS inverter and simulate it.	2
12	**Experiments 12** Measure propagation delay of CMOS inverter.	2
13	**Experiments 13** Develop the D-latch using CMOS transmission gate and simulate it	2
14	**Experiments 14** Develop the edge triggered D flip flop and simulate it.	2
Total Hours		**28**

Textbook :

1 CMOS Digital Integrated Circuits Analysis and Design, Sung Mo Kang, Tata McGraw Hill, 2003

2 CMOS VLSI Design, Neil H.E.Weste, Pearson, 2005

References:

1 Basic VLSI Design, Basic VLSI Design, Douglas A. Pucknell, Kamran Eshraghian, Prentice Hall of India, 3rdEdition, 2009

2 Introduction to VLSI circuits and systems, Introduction to VLSI circuits and systems, John P. Uyemura,, Wiley, 2rd Edition, 2014

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery

Remember / Knowledge	Understand	Apply	Analyze	Evaluate	Higher order Thinking / Creative
15.00	15.00	40.00	10.00	10.00	10.00

Instructional Method:

1 Course delivery is combined use of white board/black board and power point presentation using projector as per requirement.

2 Students will be assessed with continuous evaluation approach with assignments and quizzes at regular interval.

3 For effective communication with students, canvas LMS (Learning Management System) is used.

Supplementary Resources:

1 https://nptel.ac.in/courses/108/107/108107129

2 https://nptel.ac.in/courses/117/101/117101058/

3 https://nptel.ac.in/courses/117/101/117101004

4 http://ece-research.unm.edu/jimp/vlsi

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